

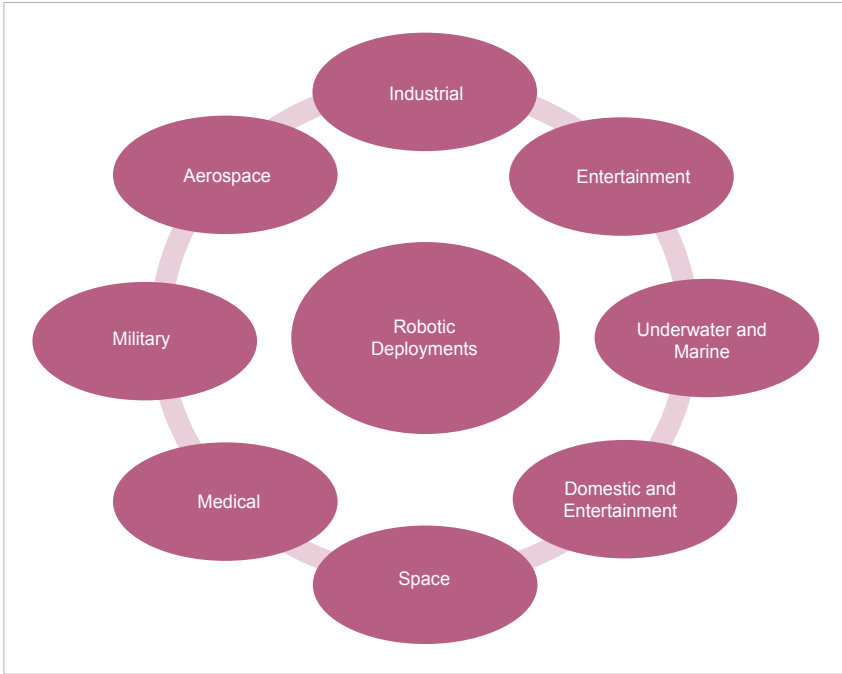


Figure 1: Key industries for robotic deployments

their work and predict results. Once the results on simulations and robotic platforms are found suitable, further actions can be planned.

Challenges that appear simple to humans differ hugely between instances of robotic activities and conditions. Such robotic systems help a lot in the sensitive environments where automated machines or programmable devices are placed. Scientists and data analysts can fetch the data via satellite or assorted wireless protocols from a remote control room.

- The typical features of robotic platforms include:
- Importing templates from CAD for mechanical and production engineers
 - Integration of OpenStreetMap and Google Maps for real-time locations
 - Easy interface to work
 - Plugins and icons for different automations including automobiles, space, drones, aerospace vehicles, volcanoes, oceans, aircraft and many others
 - Integration with multiple programming languages for

- customisation of features
 - Exporting the results in multiple formats
 - Association with multiple libraries of different engineering domains for compatibility
- Table 1 lists the free and open source robotics development frameworks that are being widely used for industrial, social, corporate, military and related applications. These are programmable libraries

Robotic simulation platform	URL
Webots	cyberbotics.com
Gazebo	gazebo-sim.org
Robot Operating System	ros.org
Open Robot Control Software	orocos.org
Yet Another Robot Platform	yarp.it/git-master
Mobile Robot Programming Toolkit	mrpt.org
Robotarium	robotarium.gatech.edu
Poppy-Project	poppy-project.org
CoppeliaSim	coppeliarobotics.com

Table 1: Free and open source platforms for robotic simulations

that can be customised as per the requirements of the domain in which they are implemented.

Using CoppeliaSim for industrial and corporate applications

CoppeliaSim is a cross-platform tool for simulation and industrial robotic applications. This tool makes it possible to create portable, scalable and easy-to-maintain simulations for multiple scenarios.

This framework has interfaces to various programming languages like C, Lua, Java, Python, Matlab and Octave. For an interactive production environment, the CoppeliaSim Robot Simulator relies on a distributed control architectural system in which the individual control of each object or model requires an embedded script for high performance applications. Such features give CoppeliaSim extremely versatile and adaptable powers for a broad range of robotic systems.

CoppeliaSim is used in the development of high-speed algorithms, as well as for plant modelling, rapid prototyping, validations, automated double-checking, etc. This feature-rich tool performs very effectively for a range of applications in domains that include engineering, military, aerospace, and health sciences.